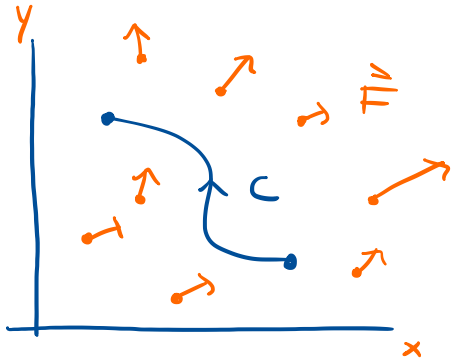
→ Do problems 1 and 2.

Let C be a curve with parameterization $\vec{r}(t)$. We think of the vector line element $d\vec{r}$ as being an infinitesimal displacement along C. If $\vec{F}$ is a vector field, the line integral of $\vec{F}$ over C is



$$\int_C \vec{F}\cdot d\vec{r}$$

It is computed using the formula

$$\int_C \vec{F}\cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t))\cdot \vec{r}\,'(t)\,dt$$

where $[a,b]$ is the parameter interval.

Fact: Any other orientation-preserving parameterization of C produces the same integral. An orientation-reversing parameterization changes the sign.

→ Do problem 3.

We can think of

$$d\vec{r} = dx\,\vec{i} + dy\,\vec{j}$$

so that if $\vec{F} = F_1\vec{i} + F_2\vec{j}$, then

$$\int_C \vec{F}\cdot d\vec{r} = \int_C \underbrace{F_1\,dx + F_2\,dy}_{\text{this is called a 1-form.}}.$$

→ Do problem 4.